Computer Organization And Architecture

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Learning Outcome

- Floating Point Representation
- IEEE 754 Standards For Floating Point Representation
- Single Precision
- Double Precision
- Single Precision Addition

Floating Point Representation

The floating point representation does not reserve any specific number of bits for the integer part or the fractional part. Instead it reserve a certain point for the number and a certain number of bit where within that number the decimal place sits called the exponent.

IEEE 754 Floating point representation

According to IEEE754 standard, the floating point number is represented in following ways:

- Half Precision(16bit):1 sign bit,5 bit exponent & 10 bit mantissa
- Single Precision(32bit):1 sign bit,8 bit exponent & 23 bit mantissa
- Double Precision(64bit):1 sign bit,11 bit exponent & 52bit mantissa
- Extend precision(128bit):1 sign bit,15bit exponent & 112 bit mantissa

Floating Point Representation

The floating point representation has two part : the one signed part called the mantissa and other called the exponent.

Sign Bit	Exponent	Mantissa

$$(\text{sign}) \times \text{mantissa} \times 2^{\text{exponent}}$$

Decimal To Binary Conversion

$(55.35)_{10} = (?)_2$

32	16	8	4	2	1
1	1	0	1	1	1

0.35×2	0	.7
0.7×2	1	.4
$.4 \times 2$	0	.8
$.8\times 2$	1	.6
$.6 \times 2$	1	.2
$.2\times 2$	0	.4

$(55)_{10}=(110111)_2$

$(0.35)_{10} = (010110)_2$

$(45.45)_{10}=(110111.010110)_2$

Scientific Notation

$$\underbrace{-}_{\text{sign}} \quad \underbrace{1.602}_{\text{significand}} \times \underbrace{10}_{\text{Base}}{}^{\overbrace{-19}^{\text{Exponent}}}$$

IEEE 32-bit floating point representation

Sign Bit	Biased Exponent	Trailing Significand bit or Mantissa

1-bit 8 -bit 23- bit

Number representation: $(-1)^S \times 1.M \times 2^{E-127}$

IEEE 32-bit floating point representation

$(45.45)_{10}=(101101.011100)_2$

Step -1: Normalize the number

Step-2: Take the exponent and mantissa.

Step-3:Find. the bias exponent by adding 127

Step-3:Normalize the mantissa by adding 1.

Step -4:Set the sign bit 0 if positive otherwise 1 .

For n bit exponent bias is $2^{n-1}-1$

IEEE 32-bit floating point representation

$(45.45)_{10} = (?)_2$

32	16	8	4	2	1
1	0	1	1	0	1

$(45)_{10}=(101101)_2$

$(0.45)_{10} = (011100)_2$

$(45.45)_{10}=(101101.011100)_2$

0.45×2	0	.9
0.9×2	1	.8
$.8 \times 2$	1	.6
$.6 \times 2$	1	.2
$.2 \times 2$	0	.4
$.4 \times 2$	0	.8

IEEE 32-bit floating point representation

$(45.45)_{10}=(101101.011100)_2$

$101101.011100 = 1.01101011100 \times 2^5$

Here bias exponent = 5 + 127 = 132

mantissa=01101011100

Sign Bit	Biased Exponent	Trailling Significand bit or Mantissa

1-bit 8 -bit 23- bit

IEEE 32-bit floating point representation

$(132)_{10}=(?)_2$

128	64	32	16	8	4	2	1
1	0	0	0	0	1	0	0

$(132)_{10}=(10000100)_2$

0	10000100	01101011100110011001100
1-bit	8 -bit	23- bit

IEEE 64-bit floating point representation

Sign Bit	Biased Exponent	Trailling Significand bit or Mantissa

1bit	11bits	52bits

Here we use $2^{11-1} - 1 = 1023$ as bias value.

IEEE 64-bit floating point representation

$(45.45)_{10}=(101101.011100)_2$

$101101.011100 = 1.01101011100 \times 2^5$

Here bias exponent $= 5 + 1023=1028= (10000000100)_2$

mantissa=$01101011\underbrace{100}$

0	10000000100	01101011100110011001100……
1-bit	11 -bits	52- bits

Convert Floating Point To Decimal

$$\underbrace{0100\ 0000\ 0}_{\text{exponent}}\underbrace{100\ 0110\ 1011\ 0000\ 0000\ 0000}_{\text{Mantissa}}$$

Number representation: $(-1)^S \times 1.M \times 2^{E-127}$

$S=0$

$E=(1000000)2=(64)_{10}$

$M =(.100\ 0110\ 1011\ 0000\ \ 0000\ \ 0000\ )_2= (0.5537109375)_{10}$

$(-1)^0 \times 1.5537109375 \times 2^{\,64-127} = 1.68453677\times10^{-19}$

Addition of floating point

First consider addition in base 10 if exponent is the same the just add the significand

5.0E+2

+7.0E+2

12.0E+2=1.2E+3

Addition of floating point

1.2232E+3 + 4.211E+5

First Normalize to higher exponent

a. Find the difference between exponents

b. Shift smaller number right by that amount

1.2232E+3=.012232E+5

Addition of floating point

4.211 E+5

+ 0.012232 E+5

4.223232 E+5

32Bit floating point addition

a 0 1101 0111 111 0011 1010 0000 1100 0011

b 0 1101 0111 000 1110 0101 1111 0001 1100

Find the 32 bit floating point number representation of a+b .

Here,

$e=(11010111)=(215)_{10}$

m= (111 0011 1010 0000 1100 0011)

32Bit floating point addition

$a= (-1)^0 \times 1.111\ 0011\ 1010\ 0000\ 1100\ 0011 \times 2^{127-215}$

$=1.111\ 0011\ 1010\ 0000\ 1100\ 0011 \times 2^{12}$

$e=(11010111)= (215)_{10}$

$m= 000\ 1110\ 0101\ 1111\ 0001\ 1100$

$b= 1.000\ 1110\ 0101\ 1111\ 0001\ 1100 \times 2^{12}$

$+\ a= \quad 1.111\ 0011\ 1010\ 0000\ 1100\ 0011 \times 2^{12}$

$11\ .\ 000\ 0\ 001\ 1111\ 1111\ 1101\ 1111 \times 2^{12}$